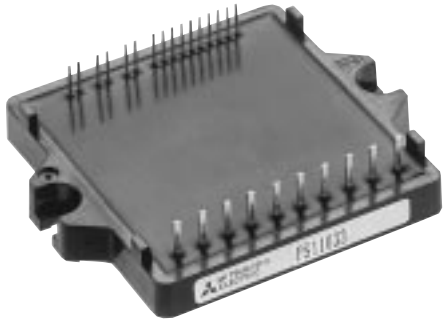
PS11033

FLAT-BASE TYPE
INSULATED TYPE

PS11033



INTEGRATED FUNCTIONS AND FEATURES

- Converter bridge for 3 phase AC-to-DC power conversion.
- 3 phase IGBT inverter bridge configured by the latest 3rd. generation IGBT and diode technology.
- Inverter output current capability I_O (Note 1):

Type Name	Motor Rating	I_O (100%)	I_O (150%; 60sec)
PS11033	0.4 kW/200V AC	3.0Arms	4.5Arms

(Note 1) : The inverter output current is assumed to be sinusoidal and the peak current value of each of the above loading cases is defined as : $I_{OP} = I_O \times \sqrt{2}$, $T_C < 100°C$

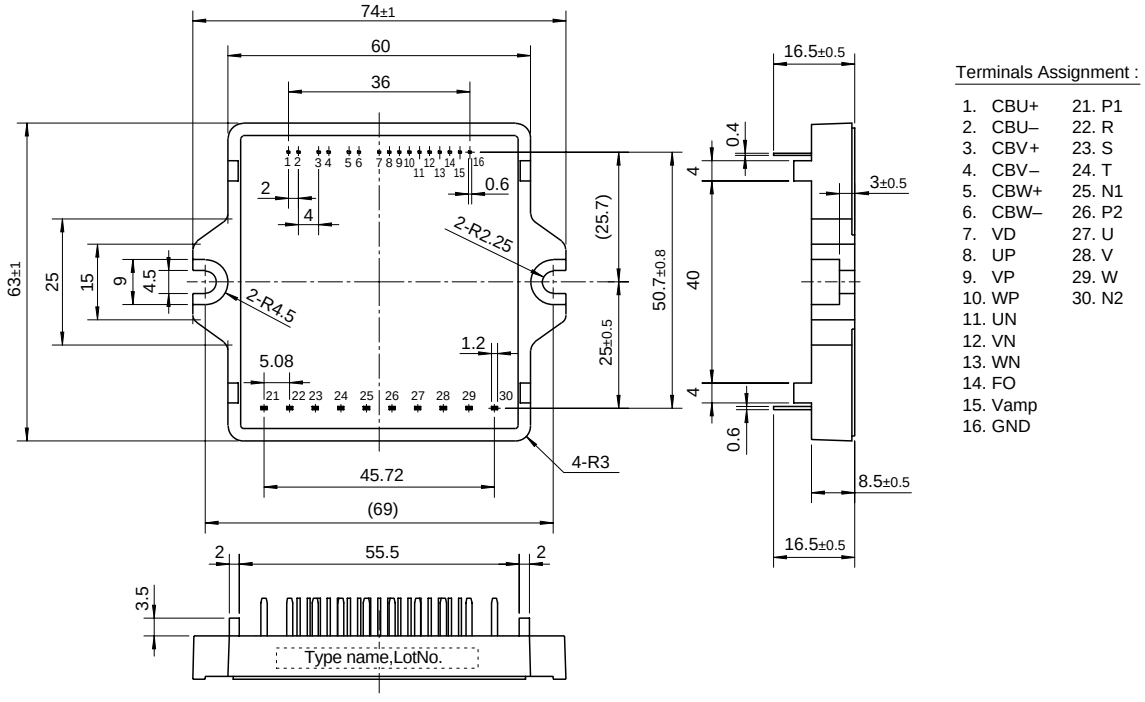
INTEGRATED DRIVE, PROTECTION AND SYSTEM CONTROL FUNCTIONS:

- P-Side IGBTs : Drive circuit, high-level-shift circuit, bootstrap circuit supply scheme for Single Control-Power-Source drive, and under voltage (UV) protection.
- N-Side IGBTs : Drive circuit, DC-Link current sense and amplifier circuits for overcurrent protection, control-supply under-voltage protection (UV), and fault output (F_O) signaling circuit.
- Fault Output : N-side IGBT short circuit (SC), over-current (OC), and control supply under-voltage (UV).
- Inverter Analog Current Sense : N-Side IGBT DC-Link Current Sense.
- Input Interface : 5V CMOS/TTL compatible, Schmitt Trigger input, and Arm-Shoot-Through interlock protective function.

APPLICATION

Acoustic noise-less 0.4kW/200V AC Class 3 phase inverters, motor control applications, and motors with built-in small size inverter package

PACKAGE OUTLINES



Terminals Assignment :

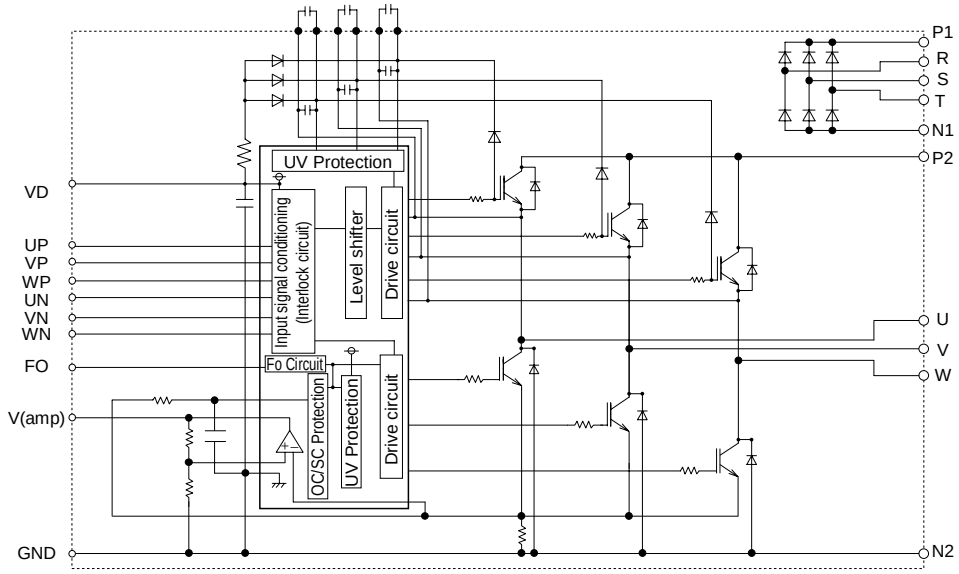
1.	CBU+	21.	P1
2.	CBU–	22.	R
3.	CBV+	23.	S
4.	CBV–	24.	T
5.	CBW+	25.	N1
6.	CBW–	26.	P2
7.	VD	27.	U
8.	UP	28.	V
9.	VP	29.	W
10.	WP	30.	N2
11.	UN		
12.	VN		
13.	WN		
14.	FO		
15.	Vamp		
16.	GND		

(Fig. 1)

PS11033

FLAT-BASE TYPE
INSULATED TYPE

INTERNAL FUNCTIONS BLOCK DIAGRAM



(Fig. 2)

MAXIMUM RATINGS (Tj = 25°C)

INVERTER PART

Symbol	Item	Condition	Ratings	Unit
V_{CC}	Supply voltage	Applied between P2-N2	450	V
$V_{CC(surge)}$	Supply voltage (surge)	Applied between P2-N2, Surge-value	500	V
V_P or V_N	Each output IGBT collector-emitter static voltage	Applied between P2-U.V.W, U.V.W-N2	600	V
$V_{P(S)}$ or $V_{N(S)}$	Each output IGBT collector-emitter switching voltage	Applied between P2-U.V.W, U.V.W-N2	600	V
±Ic(±Icp)	Each output IGBT collector current	Tc = 25°C, "( )" means Ic peak value	±8 (±16)	A

CONVERTER PART

Symbol	Item	Condition	Ratings	Unit
V_{RRM}	Repetitive peak reverse voltage		800	V
Ea	Recommended AC input voltage		220	Vrms
I_O	DC output current	3ϕ rectifying circuit	10	A
I_{FSM}	Surge (non-repetitive) forward current	1 cycle at 60Hz, peak value non-repetitive	100	A
I^2t	I^2t for fusing	Value for one cycle of surge current	42	A^2s

CONTROL PART

Symbol	Item	Ratings	Unit
V_D, V_{DB}	Supply voltage	–0.5 ~ 20	V
V_{CIN}	Input signal voltage	–0.5 ~ +7.5	V
V_{FO}	Fault output supply voltage	–0.5 ~ +7.5	V
I_{FO}	Fault output current	15	mA
I_{amp}	DC-Link IGBT current signal Amp output current	1	mA

PS11033

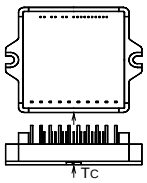
FLAT-BASE TYPE
INSULATED TYPE

TOTAL SYSTEM

Symbol	Item	Condition	Ratings	Unit
T_j	Junction temperature	(Note 2)	–20 ~ +125	°C
T_{stg}	Storage temperature	—	–40 ~ +125	°C
T_C	Module case operating temperature	(Fig. 3)	–20 ~ +100	°C
V_{ISO}	Isolation voltage	60 Hz sinusoidal AC applied between all terminals and the base plate for 1 minute.	2500	Vrms
—	Mounting torque	Mounting screw: M4	0.98 ~ 1.47	N·m

(Note 2) : The indicated values are specified considering the safe operation of all the parts within the ASIPM. The max. ratings for the ASIPM power chips (IGBT & FWDi) is Tj < 150.

CASE TEMPERATURE MEASUREMENT POINT



(Fig. 3)

THERMAL RESISTANCE

Symbol	Item	Condition	Ratings			Unit
			Min.	Typ.	Max.	
$R_{th(jc)Q}$	Junction to case Thermal Resistance	Inverter IGBT (1/6)	—	—	4.1	°C/W
$R_{th(jc)F}$		Inverter FWDi (1/6)	—	—	6.1	°C/W
$R_{th(jc)FR}$		Converter Di (1/6)	—	—	4.8	°C/W
Rth(cf)	Contact Thermal Resistance	Case to fin thermal, grease applied (1 Module)	—	—	0.074	°C/W

ELECTRICAL CHARACTERISTICS (Tj = 25°C, V_D = 15V, V_{DB} = 15V unless otherwise noted)

Symbol	Item	Condition	Ratings			Unit
			Min.	Typ.	Max.	
$V_{CE(sat)}$	Collector-emitter saturation voltage	Tj = 25°C, Input = ON, Ic = 8A, V_D = V_{DB} = 15V (Shunt voltage drop not included)	—	—	2.9	V
V_{EC}	FWDi forward voltage	Tj = 25°C, –Ic = 8A	—	—	2.9	V
V_{FR}	Converter diode voltage	Tj = 25°C, I_{FR} = 5A	—	—	1.5	V
I_{RRM}	Converter diode reverse current	V_R = V_{RRM}, Tj = 125°C	—	—	8	mA
ton	Switching times	1/2 Bridge inductive, Input = 5V ↔ 0V V_{CC} = 300V, Ic = 8A, Tj = 125°C V_D = 15V, V_{DB} = 15V Note: ton, toff include delay time of the internal control circuit.	0.3	0.6	1.5	µs
tc(on)			—	0.43	0.8	µs
toff			—	1.6	2.5	µs
tc(off)			—	0.5	1.2	µs
trr	FWDi reverse recovery time		—	0.12	—	µs
Short circuit endurance (Output, Arm, and Load Short Circuit Modes)		@V_{CC} ≤ 400V, Input = 5V → 0V (One-Shot) –20°C ≤ Tj (start) ≤ 125°C, 13.5V ≤ V_D = V_{DB} ≤ 16.5V	• No destruction • F_O output by protection operation			
Switching SOA		@V_{CC} ≤ 400V, Input = 5V ↔ 0V, Tj ≤ 125°C Ic < OC trip level, 13.5V ≤ V_D = V_{DB} ≤ 16.5V	• No destruction • No protecting operation • No F_O output			

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PS11033

FLAT-BASE TYPE
INSULATED TYPE

ELECTRICAL CHARACTERISTICS (T_j = 25°C, V_D = 15V, V_{DB} = 15V unless otherwise noted)

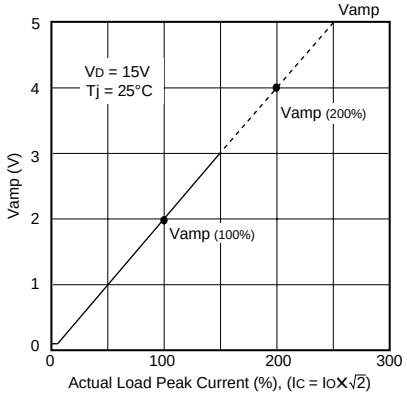
Symbol	Item		Condition		Ratings			Unit
					Min.	Typ.	Max.	
I_D	Circuit current (Average)		T_j = 25°C, V_D = 15V, Vin = 5V		—	—	50	mA
I_{DB}	Circuit current (Average)		T_j = 25°C, V_D = V_{DB} = 15V, Vin = 5V		—	—	5	mA
$V_{thon)}$	Input on threshold voltage				0.8	1.4	2.0	V
$V_{th(off)}$	Input off threshold voltage				2.5	3.0	4.0	V
R_i	Input pull-up resistor		Applied between input terminal-inside power supply		—	50	—	kΩ
f_{PWM}	PWM input frequency		T_C ≤ 100°C, T_j ≤ 125°C		1	—	15	kHz
t_{dead}	Arm shoot-through blocking time		Relates to corresponding inputs T_C = –20°C ~ +100°C (Note 3)		2.2	—	—	µs
t_{int}	Input interlock sensing		Relates to corresponding input (Fig. 6)		—	100	—	ns
Vamp(100%)	Inverter DC-Link IGBT current sense voltage output signal		I_C = $I_{OP(100\%)}$	V_D = 15V T_j = 25°C (Fig. 4)	1.5	2.0	2.5	V
Vamp(200%)			I_C = $I_{OP(200\%)}$		3.0	4.0	5.0	V
Vamp(250%)	Inverter DC-Link IGBT current sense voltage output limit		I_C = $I_{OP(250\%)}$	V_D = 15V	5.0	—	—	V
Vamp(0)			I_C = 0A	(Fig. 4)	—	50	100	mV
OC	Over current trip level		T_j = 25°C	(Fig. 5)	8.5	10.6	16.0	A
t_{OC}	Over current delay time		T_j = 25°C	(Fig. 5)	—	10	—	µs
SC	Short circuit trip level		T_j = 25°C	(Fig. 5)	—	16	—	A
t_{SC}	Short circuit delay time		T_j = 25°C	(Fig. 5)	—	2	—	µs
UV_D	Supply circuit under voltage protection	Trip level	T_C = T_j = 25°C	(Fig. 5)	11.0	12.0	13.0	V
UV_{Dr}		Reset level			11.5	12.5	13.5	V
UV_{DB}		Trip level			10.1	10.8	11.6	V
UV_{DBr}		Reset level			10.6	11.3	12.1	V
tdV		Delay time			—	10	—	µs
t_{FO}	Fault output pulse width		T_j = 25°C	(Note 4)	1.0	1.8	—	ms
$I_{Fo(H)}$	Fault output current		Open collector output	(Note 4)	—	—	1	µA
$I_{Fo(L)}$					—	—	15	mA

(Note 3) : The dead-time has to be set externally by the CPU; it is not part of the ASIPM internal functions.
(Note 4) : Fault output signaling is given only when the internal OC, SC, & UV protection circuits are activated.
The OC, SC and UV protection (and fault output) operate for the lower arms only. The OC and SC protection Fault output is given in a pulse format while that of UV protection is maintained throughout the duration of the under-voltage condition.

RECOMMENDED OPERATING CONDITIONS

Symbol	Item	Condition	Ratings			Unit
			Min.	Typ.	Max.	
V_{CC}	Supply voltage	Applied across P2-N2 terminals	—	300	400	V
V_D	Supply voltage	Applied between V_D-GND	13.5	15.0	16.5	V
V_{DB}	Supply voltage	Applied between CBU+ & CBU–, CBV+ & CBV–, CBW+ & CBW–	13.5	15.0	16.5	V
$\Delta V_D, V_{DB}$	Supply voltage ripple		–1	—	+1	V/µs
$V_{CIN(ON)}$	Input on voltage	Applied between UP • VP • WP • UN • VN • WN and GND	0	—	0.8	V
$V_{CIN(OFF)}$	Input off voltage		4.0	—	5.0	V
t_{dead}	Arm shoot-through blocking time	Relates to corresponding inputs	2.2	—	—	µs
T_C	Module case operating temperature		—	—	100	°C
f_{PWM}	PWM Input frequency	T_C ≤ 100°C, T_j ≤ 125°C	—	—	15	kHz
t_{XX}	Allowable minimum input on-pulse width		1	—	—	µs

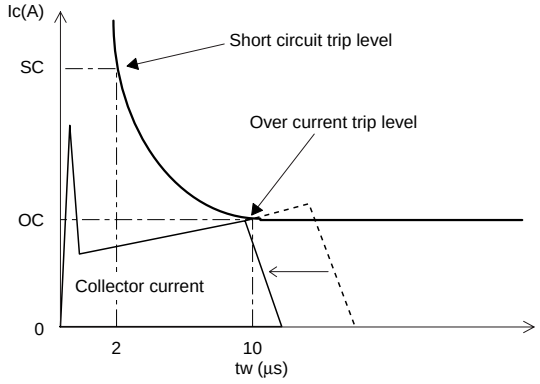
INVERTER DC-LINK IGBT CURRENT ANALOGUE SIGNALING OUTPUT (TYPICAL)



(Fig. 4)

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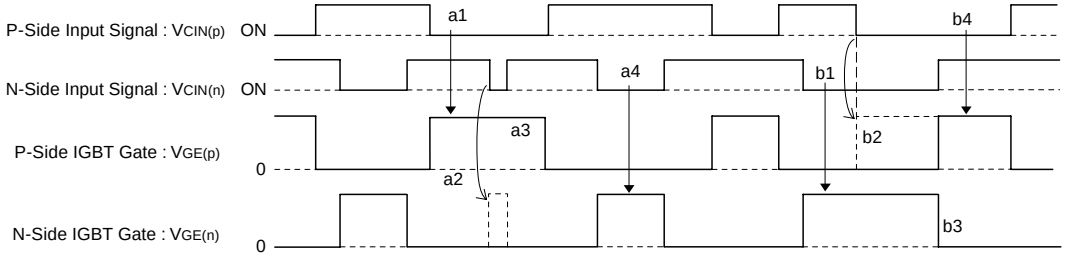
CURRENT ABNORMALITY PROTECTIVE FUNCTIONS



(Fig. 5)

Protection is achieved by monitoring and filtering the N-side DC-Bus current. The over-current protection is activated (after allowing a filtering time of 10 μs) when the line current reaches 250% of the rated load-current Io (rms). Similarly, the short circuit protection is activated (after allowing a filtering time of 2 μs) when the line current reaches twice the rated collector-current (Ic). When a current trip-level is exceeded (OC or SC), all the N-side IGBTs are intercepted (turned OFF) and a fault-signal is output. After the fault-signal output duration (1.8 ms - typ.), the interception is Reset at the following OFF input signal. However, since the fault may be repetitive, it is recommended to stop the system after the fault-signal is received and check the fault. The trip-level settings described above are summarized in the following figure:

ARM-SHOOT-THROUGH INTER-LOCK PROTECTIVE FUNCTION



(Fig. 6)

Description:

(1) During the ON-State of either of the upper-arm or the lower-arm IGBT, the inter-lock protection circuit blocks any erroneous ON pulses (resulting from input noise) from triggering the other arm IGBT and thus it prevents the arm-shoot-through situation.

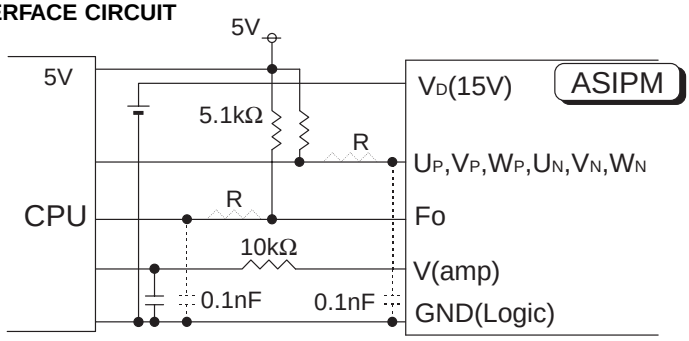
(2) When two ON-signals are received for both the upper and the lower arms, the signal received first will be passed to the IGBT and the second signal will be blocked. The second signal will be passed to its corresponding IGBT immediately after the first signal is OFF.

Note: This protective function provides no fault signaling output. The Dead-Time has to be set using the micro-controller (CPU).

Operation:

a1. P-side normal ON-signal ⇒ P-side IGBT gate turns ON.
a2. N-side erroneous ON-signal ⇒ N-side IGBT gate remains OFF.
a3. While P-side ON-signal remains ⇒ P-side IGBT gate remains ON.
a4. N-side normal ON-signal ⇒ N-side IGBT gate turns ON.
b1. N-side normal ON-signal ⇒ N-side IGBT gate turns ON.
b2. Simultaneous ON-signals ⇒ P-side IGBT gate remains OFF.
b3. N-side receives OFF-signal ⇒ N-side IGBT gate turns OFF.
b4. Immediately after (b3) ⇒ P-side IGBT gate turns ON.

RECOMMENDED I/O INTERFACE CIRCUIT



(Fig. 7)

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